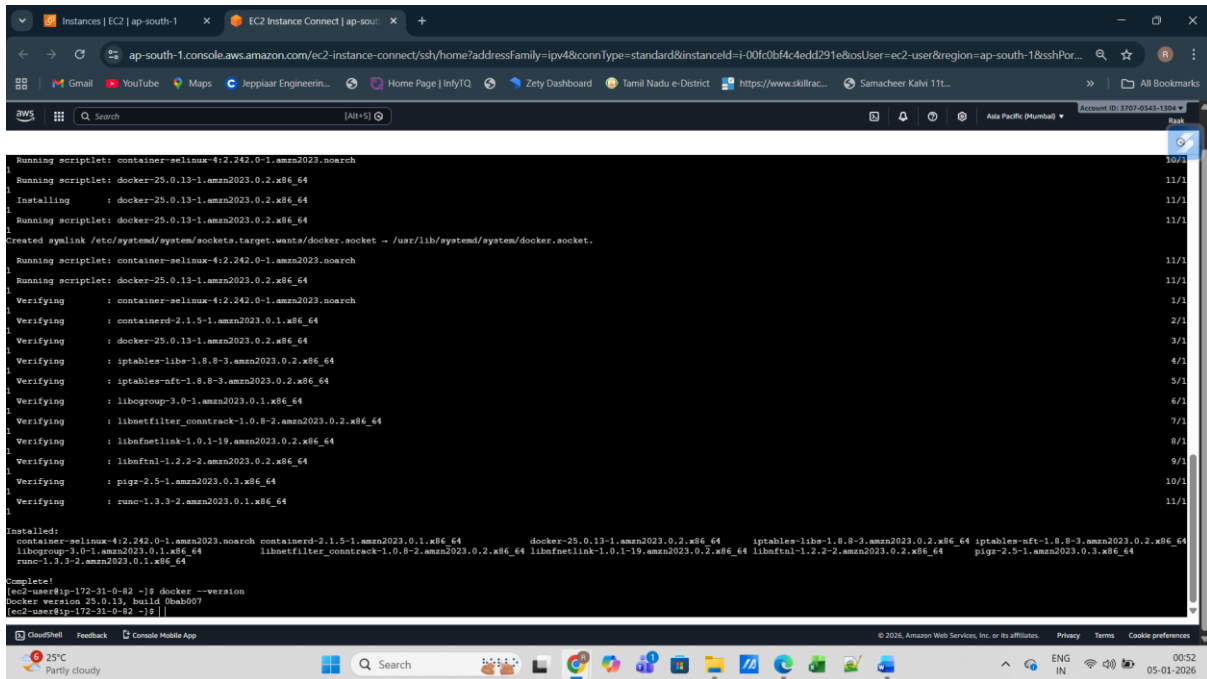


WEEK 8 ASSIGNMENT: Dockerize a Web Application

1. Installed docker and version check:



```
Running scriptlet: container-selinux-4:2.242.0-1.amzn2023.noarch
Running scriptlet: docker-25.0.13-1.amzn2023.0.2.x86_64
Installing : docker-25.0.13-1.amzn2023.0.2.x86_64
Running scriptlet: docker-25.0.13-1.amzn2023.0.2.x86_64
Created symlink /etc/systemd/system/sockets.target.wants/docker.socket -> /usr/lib/systemd/system/docker.socket.
Running scriptlet: container-selinux-4:2.242.0-1.amzn2023.noarch
Running scriptlet: docker-25.0.13-1.amzn2023.0.2.x86_64
Verifying : container-selinux-4:2.242.0-1.amzn2023.noarch
Verifying : containerd-2.1.5-1.amzn2023.0.1.x86_64
Verifying : docker-25.0.13-1.amzn2023.0.2.x86_64
Verifying : iptables-libse-1.8.8-3.amzn2023.0.2.x86_64
Verifying : iptables-nft-1.8.8-3.amzn2023.0.2.x86_64
Verifying : libhogroup-3.0-1.amzn2023.0.1.x86_64
Verifying : libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64
Verifying : libnetfilterlink-1.0.1-19.amzn2023.0.2.x86_64
Verifying : libnftnl-1.2.2-2.amzn2023.0.2.x86_64
Verifying : pigz-2.5-1.amzn2023.0.3.x86_64
Verifying : runc-1.3.3-2.amzn2023.0.1.x86_64
Installed:
container-selinux-4:2.242.0-1.amzn2023.noarch containerd-2.1.5-1.amzn2023.0.1.x86_64 docker-25.0.13-1.amzn2023.0.2.x86_64 iptables-libse-1.8.8-3.amzn2023.0.2.x86_64 iptables-nft-1.8.8-3.amzn2023.0.2.x86_64
libhogroup-3.0-1.amzn2023.0.1.x86_64 libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64 libnetfilterlink-1.0.1-19.amzn2023.0.2.x86_64 libnftnl-1.2.2-2.amzn2023.0.2.x86_64 pigz-2.5-1.amzn2023.0.3.x86_64
runcl-1.3.3-2.amzn2023.0.1.x86_64
Complete!
[ec2-user@ip-172-31-0-82 ~]$ docker --version
Docker version 25.0.13, build 0b8007
[ec2-user@ip-172-31-0-82 ~]$
```

2. Simple python code:

```
from flask import Flask
```

```
app = Flask(__name__)
```

```
@app.route("/")
```

```
def home():
```

```
    return "Hello from Dockerized Python App running on EC2 "
```

```
if __name__ == "__main__":
```

```
    app.run(host="0.0.0.0", port=5000)
```

Requirements:

Flask

3. Muti-stage build docker file:

```
# ----- Stage 1: Build -----
```

FROM python:3.10-slim AS builder

WORKDIR /app

COPY requirements.txt .

RUN `pip install --user -r requirements.txt`

----- Stage 2: Runtime -----

FROM python:3.10-slim

WORKDIR /app

```
COPY --from=builder /root/.local /root/.local
```

COPY app.py .

```
ENV PATH=/root/.local/bin:$PATH
```

EXPOSE 5000

```
CMD ["python", "app.py"]
```

Image build with below command:

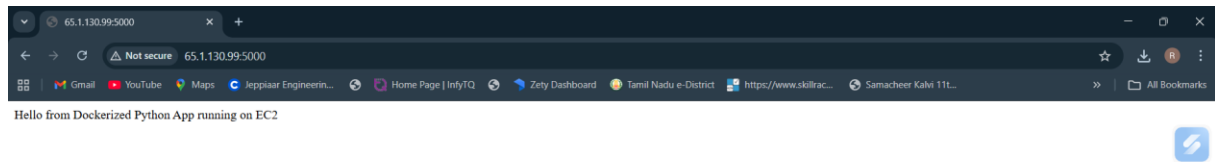
```
Docker build -t python_app .
```

[illegible]

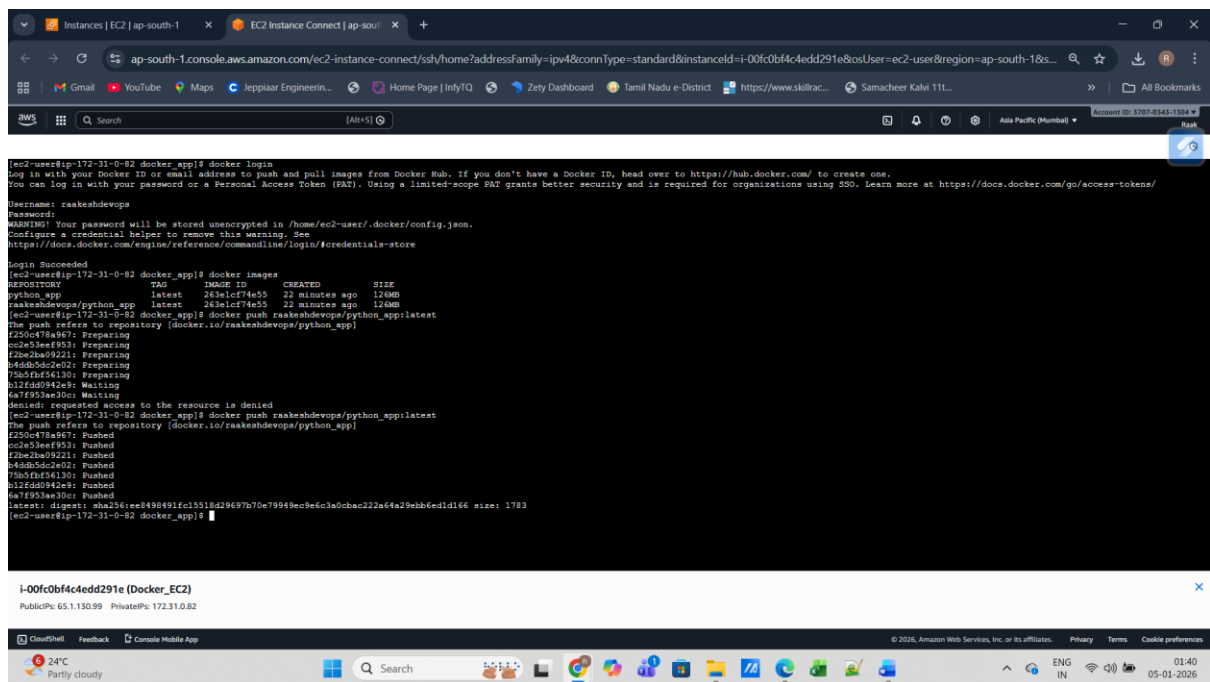
4. Running container and browser output:

```
[ec2-user@ip-172-31-0-82 docker_app]$ docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
python_app    latest    263e1cf74e55   2 minutes ago  126MB
[ec2-user@ip-172-31-0-82 docker_app]$ docker run -d -p 5000:5000 --name container1 python_app
c446b8bf8f389b96c2d45990733ee999a91baa9a4586e1e2765ff9f3fe7540da
[ec2-user@ip-172-31-0-82 docker_app]$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
c446b8bf8f38   python_app  "python app.py"         4 minutes ago  Up 4 minutes  0.0.0.0:5000->5000/tcp, :::5000->5000/tcp  container1
[ec2-user@ip-172-31-0-82 docker_app]$
```

Browser Output:



5. Pushing images to docker HUB.



Docker hub image link:

<https://hub.docker.com/repositories/raakeshdevops>

The screenshot displays the Docker Hub interface for the user 'raakeshdevops'. The page is titled 'Repositories' and shows a list of repositories within the user's namespace. The left sidebar contains navigation links for 'Repositories', 'Hardened Images', 'Collaborations', 'Settings', 'Default privacy', 'Notifications', 'Billing', 'Usage', 'Pulls', and 'Storage'. The main content area lists two repositories: 'raakeshdevops/python_app' and 'raakeshdevops/raakeshdevops'. Both repositories are public and have been pushed 4 and 5 minutes ago, respectively. The 'Contains' column shows 'IMAGE' for the first repository. The 'Scout' column indicates 'Inactive' for both. A 'Create a repository' button is visible in the top right corner of the repository list.

Name	Last Pushed	Contains	Visibility	Scout
raakeshdevops/python_app	4 minutes ago	IMAGE	Public	Inactive
raakeshdevops/raakeshdevops	5 minutes ago		Public	Inactive